Replicating Figure F1 and Equation (1) from Athey, Inoue, and Tsugawa (2025)

Jack Chepin

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1 The Original Paper

This replication is from the paper *Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance* by Susan Athey, Kosuke Inoue, and Yusuke Tsugawa, published in *AEA Papers and Proceedings* (2025).[1] The paper studies how to optimally assign treatment in settings where a randomized experiment has imperfect compliance. Imperfect compliance is when being assigned to that treatment group does not mean that one will receive the treatment. The central research question is when a program or experiment has an imperfect compliance, should it be prioritized towards those who are most likely to benefit from it, or those who are most likely to enroll, or take it (comply).

The empirical strategy is Generalized Random Forest, specifically Instrumental and Causal Forest. Causal Forest is a technique that focuses on maximizing treatment effect heterogeneity. Each tree is split such that they find what variables and cutoffs create the two groups where the treatment effects are as different from each other as possible. The Instrumental Forest is similar, but it adds the use of an Instrumental Variable. It essentially tracks heterogeneity with tree splits with local IV estimates.

2 The Result of Interest

The replication results are from the equation and table that are below:

[1] Athey, Susan, Kosuke Inoue, and Yusuke Tsugawa. 2025. "Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance." *AEA Papers and Proceedings* 115: 209–214. Supplemental Appendix available at the same URL.

1. **Equation (1)**: the Conditional Local Average Treatment Effect (CLATE), defined as:

$$\tau_{CL}(x) = \frac{\tau_Y(x)}{\tau_W(x)} \tag{1}$$

   where $\tau_Y(x)$ is the conditional Intention to Treat (ITT) and $\tau_W(x)$ is the compliance Conditional Average Treatment Effect (CATE). The paper reports a mean Local Average Treatment Effect (LATE) for debt of approximately 0.15.

2. **Figure F1**: the Targeting Operator Characteristic (TOC) curve for compliance, with a reported Area Under the Targeting Operator Characteristic curve (AUTOC) = 0.046 (SE = 0.007). This is the headline measure of how much can be gained by targeting individuals with a high estimated compliance.

These results are worth reproducing since they are central to the paper's argument. The issue of people being assigned to a treatment group, but not everyone taking it is prevalent in random experiments, including but not limited to a Medicaid lottery. Equation 1 focuses on the treatment effect for those that comply by dividing the ITT (Intention to Treat) or outcome CATE (Conditional Average Treatment Effect) (which is the expected health benefit from offering that treatment to those that have observed characteristics which are determined prior to treatment) by the conditional compliance rate (which is the subpopulation of those with observable characteristic who receive treatment if offered). The equation estimates the health benefits of the treatment given that they have observable characteristics prior and receive treatment if offered, meaning they are a complier. Figure F1 focuses on the difference between targeting high compliance individuals rather than offering Medicaid at random. The x-axis measures the percent of people that are targeted (q), and the y-axis measures the Targeting Operator Characteristic (TOC) which compares the targeted group's compliance rate to the average compliance rate, the dashed lines represent the 95% confidence bars. Both this equation and figure are essential for understanding compliance. The equation is intended to show the reduction in debt for compliers receiving the average treatment effect, and the figure shows the gain in compliance rate when targeting high-compliance individuals rather than randomly assigning it. For these impacts with respect to the objective of the paper, these results are the ones worth reproducing.

3 Data and Methods

The data came from the Oregon Health Insurance Experiment (OHIE) (Finkelstein et al. 2018),[2] which is a 2008 randomized study evaluating the causal effects of Medicaid coverage on the outcome of low-income Oregon adults. There are a total of 12,229 respondents, and of them, there were exclusions among the individuals that were missing data on treatment, age, sex, race, or education, as well as transgender individuals, yielding a final sample of $N = 12{,}208$. From this data, the instrument is lottery eligibility, the treatment Medicaid enrollment, and the outcome is whether someone doesn't have medical debt.

`code/preprocess.R` loads the raw data from `input/`, removes duplicate rows and junk columns, and imputes missing baseline covariates using the `missRanger` random forest package (as specified in Supplemental Appendix A.5). The imputed dataset is saved to `temp/`.

`code/analysis.R` reads the imputed data and estimates two GRF models. An instrumental forest (`instrumental_forest`) is utilized to estimate Equation (1), the CLATE $\tau_{CL}(x)$. A causal forest (`causal_forest`) with outcome W_i and treatment Z_i estimates the compliance CATE $\tau_W(x)$, which is then used to rank individuals for the TOC curve. The AUTOC is computed using `rank_average_treatment_effect` with doubly robust scores, following Yadlowsky et al. (2024).

The 23 pre-treatment covariates follow the paper's Supplemental Appendix A.5: age, sex, race/ethnicity, education, medical conditions (hypertension, diabetes, high cholesterol, asthma, heart attack, CHF, emphysema, kidney failure, cancer, depression), pre-lottery healthcare expenditures and ED utilization, and household size at lottery entry.

4 Replication Results

Table 1 compares our replicated estimates to the paper's reported values. Figure 1 shows the replicated TOC curve for compliance.

The replicated AUTOC of 0.0414 is extremely close to the paper's 0.046, with a difference of just -0.0046 — is well within sampling variation. The AUTOC tells us that causal forest found significant heterogeneity with respect to the compliance rate. They found that targeting higher compliance individuals rather than randomly assigned people, leads to an average of about 4.1 percentage points greater compliance. In this case, the coverage per dollar spent for those likely to enroll is more than that of those in a random assignment. As for the mean CLATE of 0.1253, it is consistent with the paper's reported overall LATE

[2]Finkelstein, Amy, Katherine Baicker, Sarah Taubman, et al. 2018. Replication Data for: "Oregon Health Insurance Experiment." Harvard Dataverse. `https://doi.org/10.7910/DVN/SJG1ED`.

Table 1: Replication of Figure F1 AUTOC from Athey, Inoue, and Tsugawa (2025)

	Replicated	Paper
AUTOC	0.0414	0.046
Standard Error	(0.0087)	(0.007)
95% CI	[0.0243, 0.0585]	[0.0279, 0.0545]
Mean CLATE	0.1253	≈ 0.15
N	12,208	

of approximately 0.15 for debt reduction among compliers. In other words, it reveals that Medicaid reduced the probability of obtaining medical debt for compliers by about 13 to 15 percentage points. Both results confirm that there is substantial, predictable heterogeneity in compliance rates across individuals, thus validating causal and instrumental forest in these circumstances. This means that the paper suggests that it is better to prioritize those who enroll in treatment rather than those who would benefit from it. When targeting by benefit, there is a failure to reject no benefit to targeting since the CLATE estimates are too noisy. Though, there is not a complete distinction. Despite the negative Spearman rank correlation between $\hat{\tau}_W(X_i)$ and $\hat{\tau}_{CL}(X_i)$ (ranking people by their estimated compliance vs benefit respectively) is -0.066 (which is a slight negative relationship), but it is a very small margin. Moreover, excluding the lowest level of compliers improves outcomes for both categories because those low-level compliers also have a low-level benefit. This is simply correlation, but there is some overlap. The number of operations (N) is also consistent with the paper's at 12,208, so, there were no observations wrongfully omitted.

The replicated standard error (0.0087) is slightly larger than the paper's (0.007), likely reflecting minute differences in the `grf` package version or number of bootstrap samples used. The error term is small, so the increase in gain from targeting is more than simply noise.

5 What I Learned

The easiest part of the replication was finding the data to replicate. The paper was available at the American Economic Association, and the data was located by link at that site, and its true location is openICPSR. The access to the data, paper, and any relevant materials such as the Supplemental Appendix made the process of replication much more efficient.

The hardest part of the replication was in fact the data preparation, specifically understanding how missing data should be handled. An initial approach using `na.omit()` reduced the sample from 12,208 to 6,816 by dropping all rows with any missing values. The paper, however, specifies `missRanger` imputation in Supplemental Appendix A.5 — a detail that

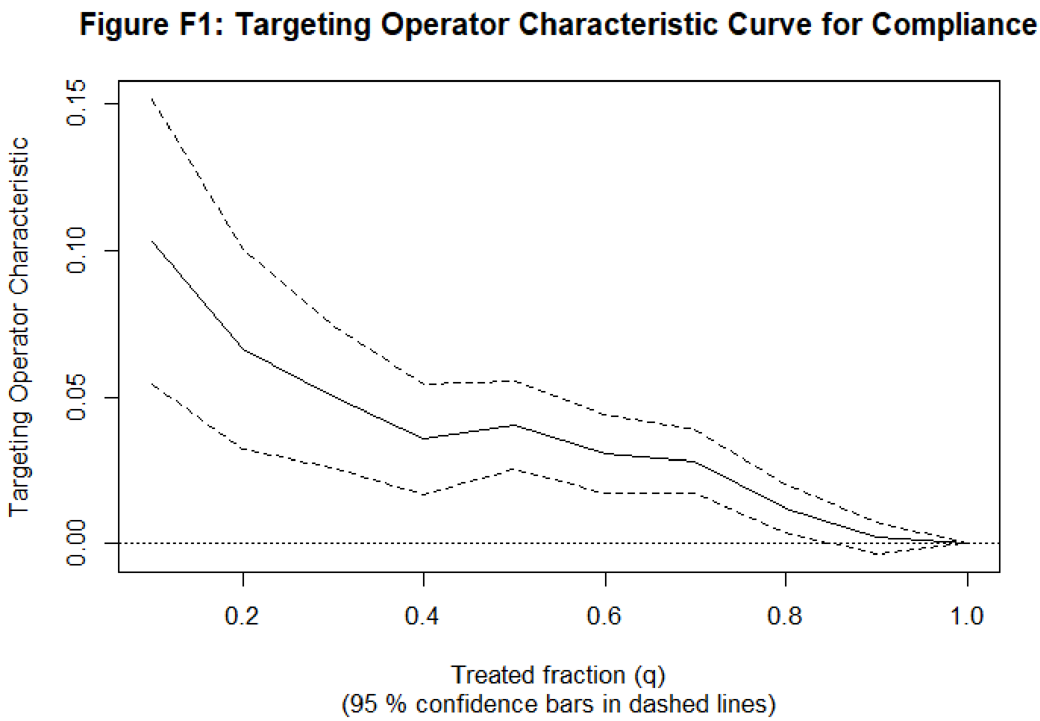

Figure 1: Replication of Figure F1: Targeting Operator Characteristic Curve for Compliance. Outcome: Compliance (W_i); Treatment: Eligibility (Z_i); Ranking rule: $r(\cdot;\ \hat{\tau}_W(X_i))$.

was essential in order to accurately obtain the results. This shows how critical it is for authors to document preprocessing decisions as carefully as estimation decisions.

A second challenge was the covariate selection. The paper describes covariates in prose rather than providing the exact variable name, this required careful cross-referencing between the dataset and the paper's description. For example, the household size variable (`numhh_list`) is a standard OHIE control but it is not explicitly listed in the paper's covariate description.

Overall, the replication taught me that GRF-based analyses are highly sensitive to sample construction. If you use the wrong command, like mentioned above, you run the risk of losing even a great portion of your sample. So you must feed the forest method correct data. If I were the original author, I would have provided an exact variable list and the preprocessing script directly in the replication package rather than describing the covariates in prose. This would reduce speculation, and potentially significant errors which in turn would ease the replication process.[3]

[3]Claude (Anthropic) was used for assistance.

References

[1] Athey, Susan, Kosuke Inoue, and Yusuke Tsugawa. 2025. "Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance." *AEA Papers and Proceedings* 115: 209–214.

[2] Finkelstein, Amy, Katherine Baicker, Sarah Taubman, et al. 2018. Replication Data for: "Oregon Health Insurance Experiment." Harvard Dataverse. https://doi.org/10.7910/DVN/SJG1ED.

[3] Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized Random Forests." *Annals of Statistics* 47(2): 1148–1178.

[4] Yadlowsky, Steve, Scott Fleming, Nigam Shah, Emma Brunskill, and Stefan Wager. 2024. "Evaluating Treatment Prioritization Rules via Rank-Weighted Average Treatment Effects." *Journal of the American Statistical Association.*